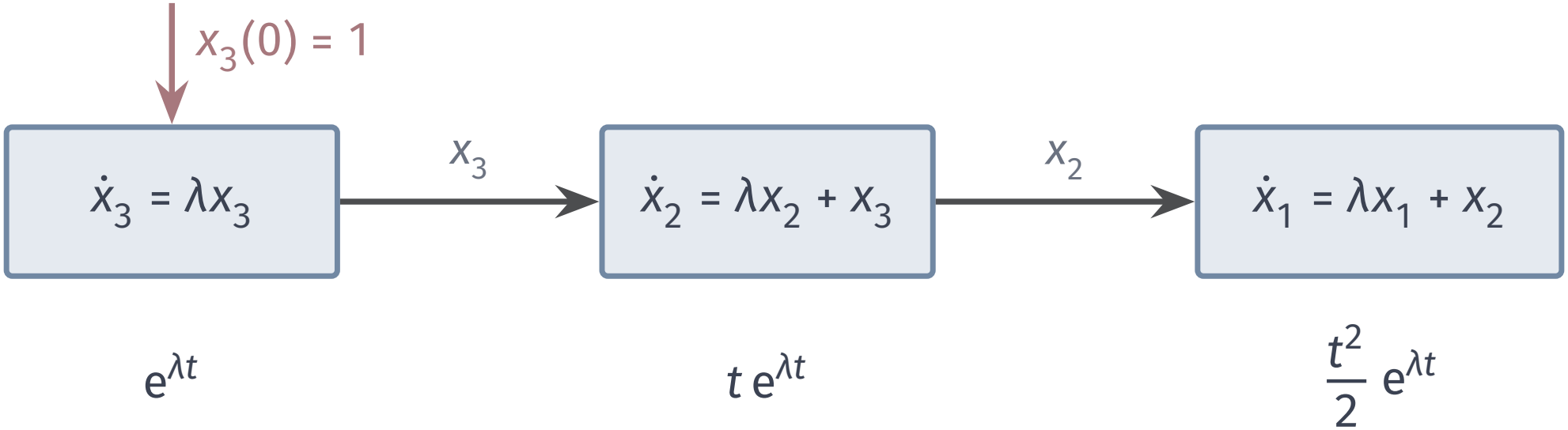
$x_3(0) = 1$
$\dot{x}_3 = \lambda x_3$
$x_3$
$\dot{x}_2 = \lambda x_2 + x_3$
$x_2$
$\dot{x}_1 = \lambda x_1 + x_2$
$e^{\lambda t}$
$t\, e^{\lambda t}$
$\frac{t^2}{2}\, e^{\lambda t}$